Lecture Scribe Tutorial 1

Tutorial_1.pdf
PDF

Tutorial_1_Solution.pdf
PDF

Generate a lecture scribe intended to serve as faithful, exam-ready reference material for revision.

Use only the provided context consisting of the lecture slides and relevant textbook excerpts. Do not introduce any material that does not appear in the given context.

While writing the scribe, reason step by step through all definitions, notation, assumptions, theorems or results, proofs or derivations, and worked example solutions exactly as presented in the context.

Make the logical flow between concepts explicit so that a student can clearly follow the reasoning during exam preparation.

Maintain a formal academic tone and structure the scribe clearly for revision purposes.

Below is a **formal, exam-ready lecture scribe** based **only** on the provided Tutorial-1 questions and official solutions .
All derivations, definitions, and logical steps follow exactly the given material.

CSE 400: Fundamentals of Probability in Computing

Tutorial-1 Lecture Scribe

(SEAS, Ahmedabad University)

Q1. Division into Equal Groups (Multinomial Counting)

Problem Structure

- 20 distinct dishes.
- Divide into 4 groups of 5 dishes each.
- Order within groups does not matter.
- Order of groups does not matter.

(a) Total Number of Ways

Since:

- All 20 dishes are distinct.
- Groups are unlabeled.
- Each group has 5 elements.

We apply the **multinomial coefficient**:

$$\text{Total ways} = \frac{20!}{(5!)^4 4!}$$

Reasoning:

- $20!$ counts all permutations.
- Divide by $5!$ for each group (order within group irrelevant).
- Divide by $4!$ because groups themselves are unlabeled.

(b) Probability All Platters Have Same Cuisine

Given:

- 5 dishes from each cuisine.
- Each cuisine must form exactly one platter.

Since platters are unlabeled, there is **exactly one favorable arrangement**.

$$P = \frac{1}{\frac{20!}{(5!)^4 4!}} = \frac{(5!)^4 4!}{20!}$$

Q2. Uniform Distribution & Waiting Time

Arrival time ~ Uniform over 30 minutes (7:00–7:30).

Key Idea:

Probability = Favorable time / Total time

(a) Wait < 5 minutes

Arrival must be in:

- 7:10–7:15
- 7:25–7:30

Total favorable time = 10 minutes.

$$P(\text{wait} < 5) = \frac{10}{30} = \frac{1}{3}$$

(b) Wait > 10 minutes

Arrival must be:

- 7:00–7:05
- 7:15–7:20

Total favorable time = 10 minutes.

$$P(\text{wait} > 10) = \frac{10}{30} = \frac{1}{3}$$

Q3. Conditional Probability with Complements

Let:

- L = arrives late
- E = leaves early

Given:

$$P(L) = 0.15, \quad P(E) = 0.25, \quad P(L \cap E) = 0.08$$

We want:

$$P(L^c \mid E^c)$$

Step 1: Complements

$$P(E^c) = 1 - 0.25 = 0.75$$

Step 2: Use Inclusion–Exclusion

$$P(L \cup E) = 0.15 + 0.25 - 0.08 = 0.32$$

Step 3: De Morgan's Law

$$P(L^c \cap E^c) = 1 - P(L \cup E) = 1 - 0.32 = 0.68$$

Step 4: Conditional Probability

$$P(L^c \mid E^c) = \frac{0.68}{0.75} = 0.907$$

Q4. Poisson Distribution (λ = 0.2)

Poisson PMF:

$$P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}$$

(a) $P(X = 0)$

$$P(X = 0) = e^{-0.2} = 0.8187$$

(b) $P(X \geqslant 2)$

$$P(X \geqslant 2) = 1 - P(0) - P(1)$$

$$P(1) = 0.2e^{-0.2} = 0.1637$$

$$P(X \geqslant 2) = 1 - 0.8187 - 0.1637 = 0.0176$$

Q5. Poisson Distribution (λ = 3.5)

(a) $P(X \geqslant 2)$

$$P(X \geqslant 2) = 1 - P(0) - P(1)$$

$$P(0) = e^{-3.5} = 0.0302$$

$$P(1) = 3.5e^{-3.5} = 0.1057$$

$$P(X \geqslant 2) = 1 - 0.0302 - 0.1057 = 0.8641$$

(b) $P(X \leqslant 1)$

$$P(X \leqslant 1) = 0.0302 + 0.1057 = 0.1359$$

Q6. Poisson (λ = 3)

(a) $P(X \geqslant 3)$

$$P(X \geqslant 3) = 1 - P(0) - P(1) - P(2)$$

$$P(0) = 0.0498$$

$$P(1) = 0.1494$$

$$P(2) = 0.2240$$

$$P(X \geqslant 3) = 1 - 0.4232 = 0.5768$$

(b) Conditional Probability

$$P(X \geqslant 3 \mid X \geqslant 1) = \frac{P(X \geqslant 3)}{P(X \geqslant 1)}$$

$$P(X \geqslant 1) = 1 - P(0) = 0.9502$$

$$= \frac{0.5768}{0.9502} = 0.607$$

Q7. Gaussian Distribution

Given PDF:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} e^{-(x+3)^2/8}$$

Compare with standard Gaussian:

$$m = -3, \quad \sigma = 2$$

Gaussian Relations

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

Results

1. $P(X \leqslant 0) = \Phi(3/2) = 1 - Q(3/2)$
2. $P(X > 4) = Q(7/2)$
3. $|X+3| < 2 \Rightarrow -5 < X < -1$

$$= \Phi(1) - \Phi(-1)$$

Using symmetry:

$$= 2\Phi(1) - 1 = 1 - 2Q(1)$$

4. $|X-2| > 1 \Rightarrow X < 1 \text{ or } X > 3$

$$= F(1) + P(X > 3) = 1 - Q(2) + Q(3)$$

Q8. Jury Decision (Conditioning)

Let:

- Each juror correct with probability θ
- Defendant guilty with probability α

If guilty:

$$\sum_{i=8}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i}$$

If innocent:

$$\sum_{i=5}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i}$$

Total probability:

$$\alpha \sum_{i=8}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i} + (1-\alpha) \sum_{i=5}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i}$$

Q9. Determining Constant c

Given:

$$p(i) = c\frac{\lambda^i}{i!}$$

Using:

$$\sum_{i=0}^{\infty} p(i) = 1$$

Since:

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

Thus:

$$ce^{\lambda} = 1$$

$$c = e^{-\lambda}$$

Results

$$P(X = 0) = e^{-\lambda}$$

$$P(X > 2) = 1 - e^{-\lambda} - \lambda e^{-\lambda} - \frac{\lambda^2}{2} e^{-\lambda}$$

CDF for Discrete Variable

$$F(a) = \sum_{x \leqslant a} p(x)$$

It is a **step function** with jumps equal to $p(x_i)$.

Q10. System Reliability (Binomial)

Each component functions with probability p.

(a) Compare 5 vs 3 Components

5-component effective if $\geqslant 3$ work:

$$10p^3(1-p)^2 + 5p^4(1-p) + p^5$$

3-component effective if $\geqslant 2$ work:

$$3p^2(1-p) + p^3$$

Difference reduces to:

$$3(p-1)^2(2p-1) > 0$$

Thus:

$$p > \frac{1}{2}$$

(b) General Case

Difference:

$$\binom{2k-1}{k} p^k (1-p)^k (2p-1)$$

Since all other terms positive:

$$> 0 \iff p > \frac{1}{2}$$

Final Conceptual Summary

This tutorial covers:

- Multinomial counting
- Uniform distribution
- Conditional probability & complements
- Poisson distribution
- Gaussian distribution & Q-function
- Law of total probability
- Discrete CDF
- Binomial reliability comparison

If you would like, I can now:

- Convert this into a **2–3 page ultra-compact revision sheet**, or
- Create a **formula + concept only quick exam sheet**, or
- Generate **exam-style tricky variations** from this tutorial.